Lecture 15

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Read Chapter 18 for next time.

Quiz 7 is in lecture next Wednesday.

Box Models

Generic Problem

- We will learn to answer questions of the following type:
  - Suppose a box contains N tickets.
  - The average of the box is μ.
  - The SD of the box is σ.
  - Suppose we draw n tickets from the box
  - We draw the tickets either with or without replacement.
- What is the chance that the sum of the n tickets is in some range (a, b)?
- What is the chance that the average of the n tickets is in some range (c, d)?

The Challenge

- Once we learn the math, answering questions of this type is easy.
- The main challenge will converting word problems to the generic problem.

Setting Up a Box Model

- Ask yourself:
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - **What is the average of the box?**
  - **What is the SD of the box?**
  - **How many tickets do we draw?**
  - **Do we draw the tickets or without replacement?**
- **Always required to know**

Example 1

- You roll a die 8 times.
- What is a box model for the sum of the rolls?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
  - How many tickets do you draw?
  - Do you draw the tickets with or without replacement?

Example 2

- You roll a die 8 times.
- What is a box model for the sum of the odd rolls?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
  - How many tickets do you draw?
  - Do you draw the tickets with or without replacement?

Example 2.5

- You roll a die 8 times.
- What is a box model for the average of the odd rolls?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
  - How many tickets do you draw?
  - Do you draw the tickets with or without replacement?

Example 3

- You roll a die 8 times.
  - If the roll comes up even, you win that many dollars.
  - If the roll comes up odd, you lose that many dollars.
- What is a box model for your net winnings?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
  - How many tickets do you draw?
  - Do you draw the tickets with or without replacement?

Example 4

- You flip a coin 8 times.
- What is a box model for the number of heads?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
  - How many tickets do you draw?
  - Do you draw the tickets with or without replacement?

Example 5

- You repeat the following 8 times:
  - You roll a die and flip a coin.
  - If the coin lands heads, you win the amount on the die.
  - If the coin lands tails, you lose the amount on the die.
- What is a box model for your net winnings?
  - How many tickets are in the box?
  - What are the numbers on the tickets?
  - What is the average of the box?
  - What is the SD of the box?
- How many tickets do you draw?
- Do you draw the tickets with or without replacement?

Additional Examples

- Additional examples available at Review_questions15.pdf

Reading Summary – Chapter 17

Section 17.1: The Expected Value

- Discussion of expected value **only in the context of box models**
- The expected value for the sum of draws with replacement is

$$\text{number of draws} \times \text{average of box}.$$

Section 17.2: The Standard Error

- Discussion of standard error **only in the context of box models**
- The standard error for the sum of draws with replacement is

$$\sqrt{\text{number of draws}} \times \text{SD of box.}$$

- The observed sum of draws is rarely more than two or three standard errors from the expected value.

Section 17.3: Using the Normal Curve

- We often can use the normal curve to approximate the chance that the sum of draws will be within some range.
- The expected value plays the role of the mean, and the standard error plays the role of the standard deviation.

Section 17.4: A Short-cut

- A short-cut formula for computing the SD of a list of numbers
- This formula applies when there are only two distinct numbers in the list
  - Example: $2, 6, 6, 2, 2, 6, 2$
- Let s be the smaller number and b the bigger number
- Let p_s and p_b be the fractions of the numbers that are s and b, respectively.
- We can show that $SD = (b - s)\sqrt{p_b * p_s}$.
- Example:

$$(6 - 2)\sqrt{\frac{3}{7} \times \frac{4}{7}}$$

Section 17.5: Classifying and Counting

- Many problems involve counting the number of times something happens
- We can usually handle this with a box model, where each ticket has either a 0 or 1.

Zero-One Boxes

- It is frequently the case that the box will only have zeros and ones.
- We refer to these as **zero-one boxes**.
- Use a zero-one box when **counting** things or when dealing with **percentages** or fractions.
- The average of a zero-one box is just the fraction of 1's in the box.

Standard Deviation of Zero-One Box

- Suppose that the box X has m 0's and n 1's.
- The average of this box is $(m * 0 + n * 1)/(m + n) = n/(m + n)$, which is the fraction of 1's in the box.
- Let's call the fraction of 1's p.
- When we square the numbers in the box, we get the same values, and so the average of the squares is also p.
- Hence $Var(X) = E(X^2) - E(X)^2 = p - p^2 = p(1 - p)$.
- Taking the square root of both sides gives $SD(X) = \sqrt{p(1 - p)}$.

Expected Value of a Single Draw

- Suppose you draw a single ticket from a box with N tickets.
  - Let X be the number you draw.
  - What is the expected value of X?
- **Expected value of a single draw = Average of the box**

Expected Value of the Sum

- Suppose we draw n tickets with replacement from a box containing N tickets.
  - Let X_1 be the first draw, X_2 be the second draw, etc.
  - Let S be the sum of the n draws: $S = \sum_{i=1}^{n} X_i$.
- $E(S) = n \times \text{average of box}$.

Without Replacement

- Same problem, but now we draw **without** replacement.
- Is $E(S) = n \times \text{average of box}$ in this case too?
  - **YES!!**
- Why?
  - Because $E(X_2) = \mu$ and $E(X_3) = \mu$ and $E(X_4) = \mu$ and so on, even when we draw without replacement.
  - Because

$$E(X_1 + \ldots + X_n) = E(X_1) + \ldots + E(X_n)$$

    no matter how $X_1, \ldots, X_n$ are related to each other.
- This assumes that $n \leq N$.

Expected Value of the Average

- Let A be the average of the n draws from the previous box:

$$A = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

- What is the expected value of A?

$$E(A) = E(1/n * S) = E(S)/n = n\mu/n = \mu$$

- $\text{expected value of average} = \text{average of box}$
  - This is always true when the draws are with replacement.
  - This is true when the draws are without replacement as long as $n \leq N$.

Variance of a Single Draw

- Suppose you draw a single ticket from a box with N tickets.
  - Let X be the number you draw.
  - What is $\mathrm{Var}(X)$?
- **Variance of a single draw = variance of the box**
- This implies that the standard error of a single draw is the standard deviation of the box.

Variance of the Sum

- Suppose we draw n tickets with replacement from a box containing N tickets.
  - Let X_1 be the first draw, X_2 be the second draw, etc.
  - Let S be the sum of the n draws: $S = \sum_{i=1}^{n} X_i$.
- What is the variance of S?
- $Var(S) = Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$
- $Var(S) = n * Var(X_1)$.
  - This is only true because the draws are all independent.
- $\text{variance of sum} = n \times \text{variance of box}$.

Standard Error of the Sum

- If we take the square root of both sides of the equation

$$\text{variance of sum} = n \times \text{variance of box},$$

- we get

$$\text{standard error of sum} = \sqrt{n} \times \text{standard deviation of box}.$$

Variance of the Average

- Let A be the average of the n draws from the previous box: $A = \frac{1}{n}\sum_{i=1}^{n} X_i$.
- What is the variance of A?
  - $Var(A) = Var(S/n) = Var(S)/n^2 = Var(X_1)/n$.
  - This only works because the draws are independent.
- $$\text{variance of average} = \frac{\text{variance of box}}{n}.$$

Standard Error of the Average

- If we take the square root of both sides of the equation $\text{variance of average} = \frac{\text{variance of box}}{n}$,
- we get

$$\text{standard error of average} = \frac{\text{standard deviation of box}}{\sqrt{n}}.$$

- This is the **square root law** and will be very important in future discussions.

Expected Value and Standard Error of a Sum from a Zero-One Box

- Suppose that we have a Zero-One Box with p the fraction of 1's.
- If we make n draws with replacement from the box, then the sum is the number of 1's in the draws.
- The expected value for the number of 1's is $n * p$ based on our previous discussion.
- The standard error for the number of 1's is

$$\sqrt{n} \times \text{standard deviation of box} = \sqrt{n * p * (1 - p)}.$$

Expected Value and Standard Error of an Average from a Zero-One Box

- If we make n draws with replacement from the box in the previous slide, then the average is the percentage of 1's in the draws.
- The expected value of this percentage is p based on our previous discussion.
- The standard error of this percentage is

$$\frac{\text{standard deviation of box}}{\sqrt{n}} = \sqrt{\frac{p(1-p)}{n}}.$$

Example

- Suppose that we make 100 draws with replacement from a box with 80% red balls and 20% green balls.
  - What is the expected value and standard error for the number of red balls?
  - What is the expected value and standard error for the percentage of red balls?
  - What is the chance of getting more than 70% red balls?
  - What would we estimate the chance as if we assume that the percentage of red balls is approximately normally distributed?

Answers

```
#E(red) = 100*0.8 = 80, SE(red) = sqrt(100*0.8*0.2)=4
#E(% red) = 0.8, SE(% red) = sqrt(0.8*0.2/100) = 0.04
#P(> 70% red) = P(> 70 red) = 1 - P(<= 70 red)
1-pbinom(70,100,0.8)
```

```
## [1] 0.988751
```

```
#Normal approximation: z = (0.705 - 0.8)/0.04 = -2.375
1-pnorm(-2.375)
```

```
## [1] 0.9912255
```

Connection with the Law of Averages

- The Law of Averages states that the absolute chance error tends to increase as the number of draws increases.
- The law also states that the absolute percentage chance error tends to decrease as the number of draws increases.
- Is this consistent with what we know about the standard error of an number or percentage from a zero-one box?

Review Questions

- See Review_questions15.pdf on Canvas.